AP PRECALCULUS

Unit 1 • Worksheet 1.12C • Answer key

Worksheet 1.12C • Transformations of Functions

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

f has domain $[0,8]$ and range $[-2,3]$. What is the range of $g(x) = -2f\left(\frac{x}{2}\right) + 1$?

(A) $[-5,5]$
(B) $[-3,7]$
(C) $[-4,6]$
(D) $[-1,3]$

Solution

Answer: (A). $-2 \cdot [-2,3] = [-6,4]$, then $+1$.

Question 2

NO CALCULATOR

For the same f, what is the domain of g?

(A) $[0,4]$
(B) $[0,16]$
(C) $[-2,6]$
(D) $[0,8]$

Solution

Answer: (B). $\frac{x}{2} \in [0,8]$.

Question 3

NO CALCULATOR

The point $(4,-2)$ is on the graph of f. Which point must be on $y = 3f(x-1) + 2$?

(A) $(3,-4)$
(B) $(5,-6)$
(C) $(5,-4)$
(D) $(4,-4)$

Solution

Answer: (C).

Question 4 (Free response) NO CALCULATOR

f has domain $[-4,2]$ and range $[-1,3]$, and $(2,3)$ is on its graph. For each function, find the domain, the range, and the image of $(2,3)$. (a) $g(x)=f(x-5)$ (b) $h(x)=-f(2x)+1$ (c) $k(x)=\frac{1}{2}f(-x)$

Solution

(a) Domain $[1,7]$, range $[-1,3]$, image $(7,3)$.
(b) Domain $[-2,1]$, range $[-2,2]$, image $(1,-2)$.
(c) Domain $[-2,4]$, range $\left[-\frac{1}{2},\frac{3}{2}\right]$, image $\left(-2,\frac{3}{2}\right)$.